

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Separates variables, at least one side correct.	AO3.1a	M1	$3\sqrt{x} \frac{dx}{dt} = 8 \sin 2t$
	Obtains correct separation PI	AO1.1b	A1	$\int 3\sqrt{x} dx = \int 8 \sin 2t dt$
	integrates 'their' expressions, at least one of 'their' sides correct	AO1.1a	M1	$\int 3x^{\frac{1}{2}} dx = \int 8 \sin 2t dt$
	Obtains correct integral (condone missing + c) CAO	AO1.1b	A1	$2x^{\frac{3}{2}} = -4 \cos 2t (+c)$
	Substitutes initial conditions, to find + c.	AO3.1b	M1	$2 \times (0)^{\frac{3}{2}} = -4 \cos(2 \times 0) + c$ $c = 4$
	Obtains a correct solution ACF	AO1.1b	A1	$x^{\frac{3}{2}} = 2 - 2 \cos 2t$
	Obtains correct solution of the form $x = f(t)$	AO2.5	A1	$x = (2 - 2 \cos 2t)^{\frac{2}{3}}$
(b)	Obtains correct max height, in cm Award FT from correct substitution into incorrect equation $x = f(t)$ but only if all three M1 marks have been awarded, must have correct units.	AO3.4	A1F	max height = $4^{\frac{2}{3}} = 252 \text{ cm}$
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving	AO3.3	M1	$\frac{dA}{dt} \propto A$

	$\frac{dA}{dt}$, A and a constant of proportionality.			$\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k dt$ $\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$ $\Rightarrow A = Be^{kt}$ AG
	Separates variables	AO1.1a	M1	
	Integrates both of 'their' sides correctly	AO1.1b	A1F	
	Constructs a rigorous mathematical argument that supports use of the given model. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(i)				
(ii)	Uses $t = 20$ and $A = 0.5$ to find k	AO3.1b	M1	When $t = 20$, $A = 0.5$ $\Rightarrow 0.5 = 0.25e^{20k}$
	Finds correct value of k	AO1.1b	A1	$\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20} \ln 2$
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	$\Rightarrow A = \frac{1}{4} (e^{\ln 2})^{\frac{t}{20}}$
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation throughout. AG	AO2.1	R1	$\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ AG
(iii)	Uses the model to set up correct equation and attempt to find t	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$ $t = 93.03$ days
	Finds correct value of t	AO1.1b	A1	
(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model	AO3.5c	E1	Introduce a limiting factor such as fish eating weed

that is specified in terms of the pond, area, weed, rate of change or time			or rate of growth decreases as surface area covered
Total 13 marks			

Q3.

(a) $\int t \cos\left(\frac{\pi}{4}t\right) dt$

Clear attempt to use parts

$$u = t \quad \frac{dv}{dt} = \cos\left(\frac{\pi}{4}t\right)$$

$$\frac{du}{dt} = 1 \quad v = k \sin\left(\frac{\pi}{4}t\right)$$

M1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) - \frac{4}{\pi} \int \sin\left(\frac{\pi}{4}t\right) (dt)$$

Must be in terms of π

A1

$$= pt \sin\left(\frac{\pi}{4}t\right) + q \cos\left(\frac{\pi}{4}t\right)$$

Correct form, any non-zero values for p, q

m1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) + \frac{4}{\pi} \times \frac{4}{\pi} \cos\left(\frac{\pi}{4}t\right)$$

Any correct unsimplified form.

Constant not required

A1

4

(b) $\int 32x \, dx = \int t \cos\left(\frac{\pi}{4}t\right) dt$

Correct separation and notation.

B1

$$16x^2 =$$

$\frac{x^2}{2}$ if 32 not brought over; allow $32 \times \frac{x^2}{2}$

B1

$$t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) + \frac{16}{\pi^2} \cos\left(\frac{\pi}{4}t\right) + C$$

Equate to result from part (a) with constant and use (0,4) to find a value for the constant

M1

$$C = 256 - \frac{16}{\pi^2}$$

Accept $C = 254$ or better(254.37886...)

A1

$$t = 45$$

$$16x^2 = -40.514... - 1.146... + 254.378... = 212.718...$$

$$x^2 = 13.294...$$

$$x = 3.646... = 3.65 \text{ m or (height =) } 365 \text{ cm}$$

$$\text{Substitute } t = 45 \text{ into } kx^2 = pt \sin\left(\frac{\pi}{4}t\right) + q \cos\left(\frac{\pi}{4}t\right) + C$$

$p \neq 0$, $q \neq 0$ and calculate x .

CSO

m1A1

6

[10]

Q4.

(a) $\frac{dh}{dt}$

B1

$$\text{derivative} = * \times (2 - h)$$

Use of $2 - h$ or $h - 2$;

* is a constant or expression in h and / or t .

M1

$$\frac{dh}{dt} = k(2 - h)$$

All correct; must be $(2 - h)$

A1

3

(b) (i) $\int x\sqrt{2x-1} \, dx = \int \frac{1}{15} dt$

*Correct separation and notation;
condone missing integral signs.*

B1

$$= \frac{1}{15} t$$

B1

Substitute $u = 2x - 1$

$$\int x\sqrt{2x-1} \, dx = \int \frac{1}{2}(u+1)\sqrt{u} \frac{1}{2} du$$

*Suitable substitution and attempt to write integral in terms
of u including dx replaced by $\frac{1}{2} du$ or $2du$.*

M1

$$= \left(\frac{1}{4}\right) \int u^{\frac{3}{2}} + u^{\frac{1}{2}} du$$

$\frac{1}{4}$ need not be seen

A1

$$= \frac{1}{4} \left(\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right) (+ C)$$

Integration correct including $\frac{1}{4}$

A1

$$x = 1; t = 0$$

$$u = 1, t = 0 \quad \frac{1}{4} \left(\frac{2}{5} + \frac{2}{3} \right) + C = 0$$

*Use $x = 1, t = 0$ to find a value for constant C from
equation in x and t .*

M1

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

A1

$$t = \frac{1}{2} \left(3(2x-1)^{\frac{5}{2}} + 5(2x-1)^{\frac{3}{2}} \right) - 4$$

$$ISW \ t = (2x - 1)^{\frac{3}{2}} (3x + 1) - 4$$

A1

Alternative (Parts)

As before

(B1B1)

$$u = x, \quad \frac{dv}{dx} = (2x-1)^{\frac{1}{2}}$$

Attempt to use parts

(M1)

$$du = 1 \quad v = k(2x-1)^{\frac{3}{2}}$$

$$\int x\sqrt{2x-1} \, dx = x \frac{1}{3}(2x-1)^{\frac{3}{2}} - \int \frac{1}{3}(2x-1)^{\frac{3}{2}} \, dx$$

Condone missing dx

(A1)

$$= x \frac{1}{3}(2x-1)^{\frac{3}{2}} - \frac{1}{15}(2x-1)^{\frac{5}{2}} + C$$

(A1)

$$x = 1, \ t = 0 \quad \frac{1}{3} - \frac{1}{15} + C = 0$$

Use $x = 1$, $t = 0$ to find a value for constant C from equation in x and t

(M1)

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

(A1)

$$t = 5x(2x-1)^{\frac{3}{2}} - (2x-1)^{\frac{5}{2}} - 4$$

$$ISW t = (2x - 1)^{\frac{3}{2}} (3x + 1) - 4$$

(A1)

8

- (ii) $x = 2 \quad t = 32.4$ (minutes)
 32.4 or better (32.373...)

B1

1

[12]

Q5.

(a) (i) $\int \frac{dx}{\sqrt{x}} = \int \sin\left(\frac{t}{2}\right) dt$

Correct separation; condone missing integral signs.

B1

$$2\sqrt{x} = -2\cos\left(\frac{t}{2}\right) (+ k)$$

$$p\sqrt{x} = q\cos\left(\frac{t}{2}\right)$$

Condone missing + k

M1

$$x = \left(-\cos\left(\frac{t}{2}\right) + C\right)^2$$

Must have previous line correct

A1

3

- (ii) $(1, 0) \quad 2 = -2 + k \text{ or } 1 = (-1 + C)^2$
Use (1, 0) to find a constant

M1

$$k = 4 \text{ or } C = 2$$

ft on $C = p - q$ from (a)(i)

A1ft

$$x = \left(2 - \cos\left(\frac{t}{2}\right) \right)^2$$

cso applies to (a)(ii)

A1

3

- (b) (i) Greatest height when $\cos(bt) = -1$

M1

Greatest height = 9 (m)

ft is (their $a + 1$)²

A1ft

2

(ii) $\cos\left(\frac{t}{2}\right) = 2 - \sqrt{5}$

$$\cos bt = a - \sqrt{5}$$

M1

$$t = 2 \cos^{-1}(2 - \sqrt{5}) = 3.6 \text{ (seconds 1dp)}$$

condone 3.6 or better (3.618.....)

A1

2

[10]

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Q6.

(a) $\frac{dA}{dt}$

B1

$$= -k$$

B1

2

- (b) (i) $A = -kt (+ C)$
Integrate

M1

$$C = 4\pi \times 60^2$$

C correct from $A = \pm k + C$

A1

$$4\pi \times 30^2 = -9k + 4\pi \times 60^2$$

Use $r = 30$ $t = 9$ and attempt to find k ,
as far as $k = \dots$

$$k = 1200\pi$$

m1

$$\begin{aligned} A &= -1200\pi t + 14400\pi \\ &= 1200\pi(12 - t) \end{aligned}$$

AG CSO

A1

4

(ii) $t = 12$ (days)

B1

1

[7]

Q7.

(a) $\int \frac{dx}{\sqrt{x+1}} = \int -\frac{1}{5} dx$

$$\frac{dt}{dx} = -5(x+1)^{\frac{1}{2}}$$

Correct separation; or
Condone missing integral signs

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$$2\sqrt{x+1} = -\frac{1}{5}t \quad (+C)$$

Correct integrals; condone $\frac{\sqrt{x+1}}{\frac{1}{2}}$

B1B1

$$x = 80 \quad t = 0 \quad C = 2\sqrt{81}$$

Use $(0, 80)$ to find a constant C

M1

$$= 18$$

F on integrals if in form $\sqrt{x+1} = qt + c$

A1F

$$x = \left(9 - \frac{1}{10}t\right)^2 - 1$$

OE; CSO; x = correct expression in t

A1

6

(b) $t = 60 \quad x = f(60)$

Evaluate f(60), ie x = ... (C not required)

M1

$= 8$

CSO

A1

2

(c) (i) $\frac{dA}{dt} = kA(9 - A)$

$\frac{dA}{dt} = \text{product involving } A; k \text{ required}$

Condone terms in t

M1A1

2

(ii) $4.5 = \frac{9}{1 + 4e^{-0.09t}}$

Condone one slip in denominator

M1

$e^{-0.09t} = \frac{1}{4}$

A1

$-0.09t = \ln\left(\frac{1}{4}\right)$

Take ln correctly

m1

$$t = \frac{\ln\left(\frac{1}{4}\right)}{-0.09}$$

$$= 15.4 \text{ (hours)}$$

CAO; condone more than 3sf if correct
 15.40327068 Allow 15h 24m

A1

4

[14]

Q8.

(a) (i) $\int \frac{dx}{e^{\frac{1}{2}x}} = \int -kt \, dt$

Separate; condone missing integral signs

B1

$$-2e^{-\frac{1}{2}x} = -k \frac{t^2}{2} \quad (+C)$$

B1B1

3

(ii) $-2e^{\frac{1}{2}x} = -k \frac{t^2}{2} - 2e^{-3}$

Use (6, 0) to find constant

M1

$$\ln\left(e^{\frac{1}{2}x}\right) = \ln\left(k \frac{t^2}{4} + e^{-3}\right)$$

Take logarithms correctly; condone one side negative. Must have a constant.

M1

$$-\frac{1}{2}x = \ln\left(k \frac{t^2}{4} + e^{-3}\right)$$

$$x = -2 \ln\left(\frac{kt^2}{4} + e^{-3}\right)$$

Answer given; CSO

A1

3

(b) (i) $t = 10 \quad x = -2 \ln \left(\frac{0.004 \times 10^2}{4} + e^{-3} \right)$

M1

$$= 3.8 \Rightarrow 3800$$

CAO

A1

2

(ii) $x = 0 \quad \frac{0.004 \times t^2}{4} + e^{-3} = 1$

M1

$$t = 30.8$$

CAO

Treat 0.04 or 0.0004 as misread (– 1)

A1

2

[10]

Q9.

(a) $\int x \, dx = \int 150 \cos 2t \, dt$

*Correct separation; condone missing $\int$
signs; must see dx, dt*

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B1

$$\frac{1}{2} x^2 = 75 \sin 2t \quad (+C)$$

Correct integrals

Accept $\frac{1}{2} \times 150$

B1B1

$$\left(20, \frac{\pi}{4} \right) \quad \frac{1}{2} \times 20^2 = 75 \sin \left(2 \times \frac{\pi}{4} \right) + C$$

C present. Use $\left(20, \frac{\pi}{4} \right)$ to find C

M1

$$C = 125$$

$$F \text{ on } x^2 = k \sin 2t$$

A1F

$$x^2 = 150 \sin 2t + 250$$

Correct integrals and evaluation of C

A1

6

(b) (i) $t = 13 \quad x^2 = 150 \sin 26 + 250 (= 364.38)$

*Evaluate $x^2 = f(13)$; $x^2 = k \sin 2t + c$
with numerical k and t*

M1

$$x = 19.1 \text{ (cm)}$$

AWRT

A1

2

(ii)
$$\left. \begin{array}{l} x = 11 \quad \sin 2t = -\frac{129}{150} \quad (= -0.86) \\ \text{or} \quad 2t = -1.035 \dots, 4.176 \dots \end{array} \right\}$$

M1

$$t = 2.1 \text{ (seconds)}$$

AWRT

A1

2

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[10]

Q10.

(a) (i)
$$\int \frac{dy}{y} = \int \sin t \, dt$$

Attempt to separate and integrate

M1

$$\ln y = -\cos t + c$$

A1 for $\ln y$; A1 for $-\cos t$; condone missing c

A1,A1

$$y = ae^{-\cos t}$$

A present; or $y = e^{-\cos t + C}$

A1

4

(ii) $y = 50, t = \pi: 50 = Ae^{-\cos \pi} = Ae$

Substitute $y = 50, t = \pi$ to find constant
 Can have $50 = e^{1+C}$ if substituted in above

$$e^C = \frac{50}{e}$$

M1
A1

$$y = 50e^{-1}e^{-\cos t}$$

AG (convincingly obtained)

A1

3

Alternative:

Must have a constant in answer to (a)(i)

$$y = Ae^{-\cos t} \text{ or } y = e^{-\cos t + C} \text{ or } \ln y = -\cos t + C$$

$$50 = Ae^{-\cos \pi} \quad 50 = e^{-\cos \pi + C} \quad \ln 50 = -\cos \pi + C$$

(M1)

$$50 = Ae \quad 50 = e^{1+C} \quad \ln y = -\cos t + \ln 50 - 1$$

(A1)

$$y = 50e^{-1-\cos t} \quad y = e^{-\cos t} \frac{50}{e} \ln\left(\frac{y}{50}\right) = -1 - \cos t$$

(A1)

Alternative:

Substitute $y = 50, t = \pi$ into $\ln y = -\cos t + C$ M1

$\ln y = -\cos t + \ln 50 - 1$ A1

$$\ln \frac{y}{50} = -1 - \cos t$$

(AG)

A1

(b) (i) $t = 6: y = 50e^{-1}e^{-\cos 6} = 7.0417... \approx 7\text{cm}$

Degrees 6.8 SC1

7 or 7.0 for A1

M1A1

2

(ii) $t = \pi \Rightarrow (\sin t = 0 \Rightarrow) \frac{dy}{dt} = 0$

Condone x for t

B1

$$\frac{d^2y}{dt^2} = y \cos t + \frac{dy}{dt} \sin t$$

For attempt at product rule including $\frac{dy}{dt}$

term; must have $\frac{d^2y}{dt^2} =$

M1

$$t = \pi$$

A1

$$\frac{d^2y}{dt^2} = y \cos \pi + \frac{dy}{dt} \sin \pi$$

$$= -50 \Rightarrow \max$$

Accept $= -y$, with explanation that y is never negative

A1

4

Alternative:

$$y = 50e^{-(1+\cos t)} = \frac{50}{e} e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e} e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2y}{dt^2} = \frac{50}{e} e^{-\cos t} \times \cos t + \frac{50}{e} e^{-\cos t} \times \sin^2 t$$

Attempt at product rule

(M1)

Correct

(A1)

$$\text{Substitute } t = \pi \rightarrow -50 \Rightarrow \max$$

(A1)

[13]

Q11.

(a) $\int \frac{dx}{\sqrt{x-6}} = \int -2dt$

Attempt to separate and integrate

M1

$$2\sqrt{x-6} = -2t + c$$

c on either side

A1A1

$$t = 0 \quad x = 70 \quad \Rightarrow \quad c = 16$$

Follow on c from sensible attempt at integrals

($\sqrt{\quad}$ not $\ln$)

m1A1F

$$t = 8 - \sqrt{x-6}$$

CAO (orAEF)

A1

6

- (b) (i) The liquid level stops falling/flowing/at minimum depth

B1

1

(ii) $x = 22 \quad t = 8 - \sqrt{22-6}$

Use $x = 22$ in their equation provided there is a c

Or start again using limits

(M1) $2\sqrt{64} - 2\sqrt{16} = \pm 2t$, (A1) $t = 4$

M1

$$t = 4$$

CAO

A1

2

[9]

Q12.

(a) (i) $t = 0$ $x = 85$

B1

1

(ii) $t = 30$ $x = 15 + 70e^{-\frac{30}{40}}$

M1

$= 48.06 \approx 48^\circ\text{C}$

A1

2

(iii) $x = 60$ $\frac{60 - 15}{70} = e^{-\frac{t}{40}}$

M1

$\ln\left(\frac{45}{70}\right) = -\frac{t}{40}$
Their 60 - 15

m1

$t = 17.67 \approx 18$ minutes

A1

3

(b) (i) $\int \frac{dx}{x - 15} = -\int \frac{dt}{40}$

Attempt to separate and integrate

M1

Correct expression and used

A1

$\ln(x - 15) = -\frac{t}{40} (+c)$

A1A1

$(0, 85) \quad c = \ln 70$

m1

$$\frac{t}{40} = \ln 70 - \ln (x - 15) \quad t = 40 \ln \left(\frac{70}{x - 15} \right)$$

Use (0.85) to find c , which must now appear in expression
Manipulate to $t = \dots$

A1

6

(ii) $e^{\frac{t}{40}} = \frac{70}{x - 15}$

Manipulate expression including a $\ln$ towards $x = \dots$

M1

$$x - 15 = 70e^{-\frac{t}{40}}$$

AG; convincingly obtained

M1A1

2

[14]

Q13.

(a) Using $F = ma$:

$$-4v^{\frac{1}{3}} = 12 \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -\frac{1}{3} v^{\frac{1}{3}}$$

$$-3 \int \frac{dv}{v^{\frac{1}{3}}} = \int dt$$

condone -, 3 incorrect side

M1

$$-3 \times \frac{v^{\frac{2}{3}}}{\frac{2}{3}} = t + c$$

$$-\frac{9}{2} v^{\frac{2}{3}} = t + c$$

condone lack of $+ c$

A1

When $t = 0$, $v = 8 \Rightarrow c = -18$

M1A1

$$-\frac{9}{2}v^{\frac{2}{3}} = t + 18$$

$$v^{\frac{2}{3}} = 4 - \frac{2}{9}t$$

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

A1

6

(b) Particle is at rest when $4 - \frac{2}{9}t = 0$

The value of t is 18

B1

1

[7]

Q14.

(a) Using $F = ma$

$$1600 \frac{dv}{dt} = 4000 - 40v$$

$$\frac{dv}{dt} = \frac{4000 - 40v}{1600}$$

M1

$$\frac{dv}{dt} = \frac{100 - v}{40}$$

A1

2

(b) $40 \frac{dv}{100 - v} = dt$

B1

$$40 \int \frac{dv}{100-v} = \int dt$$

M1

$$-40 \ln(100-v) = t + c$$

Condone lack of '+ c'

A1

$$\text{When } t = 0, v = 0 \Rightarrow c = -40 \ln 100$$

$$-40 \ln(100-v) = t - 40 \ln 100$$

$$t = 40 \ln \frac{100}{100-v}$$

$$e^{\frac{t}{40}} = \frac{100}{100-v}$$

M1A1

$$v = 100 - 100 e^{-\frac{t}{40}} \text{ or } 100(1 - e^{-\frac{t}{40}})$$

A1

6

[8]

Q15.

(a) using $F = ma$

$$0.4 \frac{dv}{dt} = 2 - 4v$$

M1

$$\frac{dv}{dt} = -10(v - 0.5)$$

Needs line above

A1

2

(b) hence $\int \frac{1}{v-0.5} dv = - \int 10 dt$
 $\ln(v - 0.5) = -10t + c$

M1 for any side integrated correctly

M1A1

m1 for + c (and M1 gained)

m1

$$v - 0.5 = Ce^{-10t}$$

$$t = 0, v = 1$$

$$\therefore C = 0.5$$

A1

$$\therefore v = 0.5 + 0.5e^{-10t}$$

$$\text{condone } v = 0.5 + e^{-10t-0.693}$$

A1

5

(c) when $v = 0.55$, $0.55 = 0.5 + 0.5e^{-10t}$

substitute 0.55 into C's (b), after finding c, possible numerical error

M1

$$10 = e^{10t}$$

$$t = \ln 10 \div 10$$

A1

$$= 0.230$$

A1

3

[10]

Q16.

(a) Using $F = ma$,

$$-0.2mv^{\frac{1}{2}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

AG Must see equ'n containing m

B1

1

(b)

$$\int \frac{dv}{v^{\frac{1}{2}}} = - \int 0.2 dt$$

M1

$$2v^{\frac{1}{2}} = -0.2t + c$$

m1 for + c

A1m1

$$\text{When } t = 0, v = 16 \therefore C = 8$$

A1

$$2v^{\frac{1}{2}} = -0.2t + 8$$

$$v = (4 - 0.1t)^2$$

AG

A1

5

- (c) When $v = 1$, $1 = (4 - 0.1t)^2$
 $4 - 0.1t = \pm 1$

M1

$$t = 30 \text{ or } 50$$

$$\left[\text{if use } 2v^{\frac{1}{2}} = 8 - 0.2t \text{ no need to see } 50 \right]$$

A1

$$t = 30$$

t ≠ 50 as ball stops when t = 40

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A1

3

- (d) Integrating $v = (4 - 0.1t)^2$

$$v = 16 - 0.8t + 0.01t^2$$

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3 + d$$

$$\text{M1 for first 3 terms or } -\frac{10}{3}(4 - 0.1t)^3$$

M1

$$\text{When } t = 0, x = 0 \Rightarrow d = 0$$

A1

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3$$

When speed is 1 m s^{-1} , $t = 30$

$$x = 480 - 360 + 90$$

dep on M1 above

m1

$$= 210$$

[No 'd', 3 marks only]

A1

4

[13]

Q17.

(a) Using $F = ma$:

$$- \lambda m v^{\frac{3}{2}} = m \frac{dv}{dt}$$

M1

$$\therefore \frac{dv}{dt} = - \lambda v^{\frac{3}{2}}$$

AG

A1

2

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(b) $\int \frac{dv}{v^{\frac{3}{2}}} = -\lambda \int dt$

M1

$$- \frac{2}{v^{\frac{1}{2}}} = -\lambda t + c$$

Condone no '+c'

A1

When $t = 0$, $v = 9 \Rightarrow c = -\frac{2}{3}$

Dep. on correct integration

(accept sign or ' $\frac{1}{2}$ ' error)

M1A1

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$$

A1

$$\frac{\sqrt{v}}{2} = \frac{1}{\lambda t + \frac{2}{3}}$$

$$v = \left(\frac{6}{2 + 3\lambda t} \right)^2$$

Needs correct algebra

m1

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

AG

A1

7

(c) When $v = 4$,

$$\text{or } \frac{36}{(2 + 3\lambda t)^2} = 4 \quad \text{M1}$$

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3} \Rightarrow 1 = \lambda t + \frac{2}{3}$$

$$(2 + 3\lambda t)^2 = 9 \quad \text{A1}$$

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M1A1

$$t = \frac{1}{3\lambda}$$

$$t = \frac{1}{3\lambda} \quad \text{A1 needs statement why}$$

$$2 + 3\lambda t \neq -3$$

A1

3

[12]

Q18.

- (a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

only for fully correct solutions

M1A1

4

- (c) When $v = 10$, $10 = 20e^{-0.05t}$

$$e^{0.05t} = 2$$

M1

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9 \text{ or } 20 \ln 2$$

A1

3

- (d) Work done by force is change in KE of car

M1

$$= \frac{1}{2}m(20)^2 - \frac{1}{2}m(10)^2$$

A1

$$= 150 \text{ m}$$

A1

3

[11]

Q19.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

B1

M1

When $t = 0$, $v = 20$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9$$

Accept 20 In 2

A1

3

[8]

Q20.

(a) $20 \frac{dv}{dt} = -10\sqrt{v}$

applying Newton's second law with $\frac{dv}{dt}$

M1

correct differential equation

A1

$$\frac{dv}{dt} = -\frac{\sqrt{v}}{2}$$

separating variables

dM1

$$\int \frac{1}{\sqrt{v}} dv = \int -\frac{1}{2} dt$$

AG

$$2\sqrt{v} = \frac{t}{2} + c$$

integrating

dM1

correct integrals

A1

$$t = 0, v = 25 \Rightarrow c = 10$$

finding the constant of integration

$$v = \left(\frac{20-t}{4} \right)^2$$

correct final result from correct working

A1

7

(b) $t = 20$

correct time

B1

1

[8]

Q21.

(a) $mg - 0.1m v = m \frac{dv}{dt}$
Newton's 2nd law

M1

$$\frac{dv}{dt} = g - 0.1v$$

CAO

A1

2

(b) $\int_0^v \frac{dv}{g - 0.1v} = \int_0^t dt$
Attempt at integration with correct separation of variables

M1

$$[-10 \ln(g - 0.1v)]_0^v = [t]_0^t$$

-1 EE

A2,1

$$\ln \frac{g - 0.1v}{g} = -0.1t$$

A1F

$$v = 10g(1 - e^{-0.1t})$$

Or equivalent

A1F

5

(c) As $t \rightarrow \infty$, $e^{-0.1t} \rightarrow 0$

M1

1

$$\therefore v \rightarrow 10g = 98$$

[8]

Q22.

(a) $m \frac{dv}{dt} = -mkv^3$

Forming a differential equation

M1

$$\int \frac{1}{v^3} dv = -\int k dt$$

$$-\frac{1}{2v^2} = -kt + c$$

Integrating to get a $\frac{1}{v^2}$ term
Correct Integral including c

m1 A1

$$v = U, t = 0 \Rightarrow c = -\frac{1}{2U^2}$$

Finding c
Correct c

m1 A1

$$\frac{1}{2v^2} = kt + \frac{1}{2U^2} = \frac{2ktU^2 + 1}{2U^2}$$

$$v^2 = \frac{U^2}{2ktU^2 + 1}$$

Correct final answer from correct working

A1

6

(b) v tends to zero

Allow decreases

B1

1

[7]